

Project Information

Proje Adı / Project Name: Location-Based Time Machine

Takım Adı / Team Name: İPEN

Project Subject: An augmented reality system that allows people to experience the past of historical sites on their phones based on their locations.

Keywords: Augmented Reality, Historical Exploration, Smart Cities, Exploration, Education, Tourism

Project Summary

Our project is a mobile application for helping people experience the past environments of the historical sites. It is an AR based application. People will be able to do that by pointing their phone to the location they want to see and will be able to see the visuals of the old times. With the application, people will be able to see past events or old buildings in that area and also learn more about the history with the digital storytelling tour guide feature. The tour guide also has an AI based chatbot which will answer your questions related to the topic. The project's purpose is to enhance tourism, give people an unique experience while exploring the cities, educating more people and helping students learn more about the history in an easier way.

Novelty

Our project uses both AR visualization and AI based tour guide (historical storytelling). Unlike other AR tourism apps, our application mostly specializes with using GPS, AI based image recognition and 3D constructions of images to create and enhance the AR experience while visiting the area. Also it has an AI based tour guide option for historical storytelling and answering questions of the users.

Here is a list of our unique features:

- AR Historical Construction – The users will be able to see how the site looked in the old times.
- Time Period Slider – The users will be able to see the different time periods.
- AI Tour Guide – The users will be able to listen to the history of the site while exploring it.
- Gamification – The users will be able to collect, unlock new badges the more they listen and explore the new places. This will help use to keep the users engaged and motivated.

And if we compare our app with the other similar ones we can say unlike Google Maps or Wikipedia, our app provides AR, visual experiences instead of just text-based information. Also there are museum guides which are very limited, only special to the special museums while our app works outdoors at any historical location, while travelling through the city. Before we present these ideas to the market, we need to conduct more research on optimizing AR for mobile devices, how to partner with museums and historical archives for data gathering, and also how to develop high quality AR based reconstruction of the historical sites.

Market and Competition

Our target audience is tourists, history organizations, students and educators. We will enhance the visiting experience of the travellers with AR experience and AI tour guide, for the students our application will help them to gain informations about history in a more fun and realistic way, and also with the collaboration with museums and governmental culture organizations our app will help them to create a smart city experience along with the improved museum experiences. For the competitors, we have Google Maps (street view feature) which has no AR feature with limited historical data. When we try to find even some basic information from Google Maps while visiting the city it is mostly impossible to find something relevant about the history of the building, area etc. There is also an app called AR Time Travel which might sound similar at first sight however we provide a more detailed database and also we have an AI tour guide along with the AR visuals and they are only focused on some certain places while our app is designed for smart cities which consist of a more wide experience. And finally as we mentioned before there are museum apps, audios special to the museums which have very low coverage, they are not supported outside of the museums, even though they are developed with the help of experts, the information accessibility is limited for people. Our app is designed to reach a wider range of people. As we investigate the market we can see that the tourism market is expanding and the demand for AR technologies is increasing. The challenge for us is to make sure our data is correct to ensure a reliable service and we need to find partnerships with the cultural organisations.

Financing

The budget requirement we foresee for our project expenses is calculated as: \$30,000 for App Development & AR Integration, \$20,000 for 3D Model & Content Creation, \$10,000 for Marketing & Partnerships and \$5,000 for Server & Maintenance Cost. The expenses of the team that will work are also included in these. The funding sources we plan to provide the necessary budget and the sustainability of our project are: Cultural Ministries Funds, Premium subscription system for some special content and angel investors. We plan to keep the pricing low for national museums at the beginning and make different pricing for the international area, which we think will be the main source of income as the project expands. We plan to work with researchers working on that region to make the data more reliable and to present the interface as simple and multilingual so that the project can appeal to a wide audience.

Management

There will be 4 teams in our project. These are Developers, Historians, 3D Designers and Marketing & Business Experts. Developers will take part in AR, GPS tracking and application development. When selecting these people, we expected the criteria of expertise in AR technology and mobile development from the people. When selecting historians, we looked for the criteria of being eager to research and having knowledge about the museums and historical places that will be the first collaborations. Historians will take part in preparing detailed reports about the project regions. 3D Designers will model the past or future estimated state of the region using AI by taking the reports from historians. We expected these people to be experts in 3D modeling. People in the marketing department will collaborate with museums and institutions and work to provide the necessary financing. We expected these people to be able to collaborate with tourism boards and be experts in digital marketing. İpek Güneş Aymergen will lead the team of Developers and Mehmet Enes Tekgöz will lead the team of 3D Designers. The marketing team will be led by a person with high communication skills and an expert in the field that we have already known.